Module 06: Learning — Bugs and Updates

Learning Objectives

1. Understand that a **Bug** is just a mistake in the code.
2. Learn about **Debugging** (Being a Code Detective).
3. Know that **Updates** make things better (Version 2.0).

Introduction: The Glitch

Have you ever seen a game character get stuck in a wall? That is a **Glitch** (or a Bug). The computer didn't know what to do, so it got confused. It doesn't mean the computer is broken. It means the **Code** needs to be fixed. We learn by fixing bugs.

Key Concepts

1. The Bug (The Moth)

The first computer bug was a literal moth stuck in a machine! Now, it means a mistake in logic.

- "Walk forward 10 steps."
- (Oops, there is a cliff).
- Bug: Character walks off cliff.

2. Debugging (The Detective)

How do you fix it? You can't just yell at the computer. You have to step through the code line by line. "Aha! I forgot to tell it to stop at the cliff." Fixing the error helps the System learn.

3. Updates (Version 2.0)

Why does your tablet need to update? Because the programmers found bugs and fixed them! Every update makes the system smarter, faster, and safer. You are like an app; you are always updating too.

Activities

Activity 1: Debug the Robot

Parent acts as a Robot with a "Bug." "I will make a sandwich." (Robot puts bread on head). Child must say "Stop! Debug!" Child gives new instruction: "Put bread on plate." Robot tries again.

Activity 2: Find the Error

Write a simple sequence on paper:

1. Put on socks.
2. Put on shoes.
3. Tie laces. Now, mix them up:
4. Put on shoes.
5. Tie laces.
6. Put on socks. Ask the child: "What is the Bug?" (Socks go inside shoes!).

Summary

Mistakes are not failures. They are just bugs waiting to be debugged. Keep updating yourself!

References

- *Grace Hopper: Queen of Computer Code* by Laurie Wallmark
- *Hello Ruby* (Debugging chapter)